A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-90903207})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary ideal I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{\,(1-\sigma_x)^2}\,(t)\,J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 673 & 535 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,673 + \mathbf{Z}\,(535 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 22725802 = 0$, of discriminant -90903207. Class group invariants $(110 \quad 10 \quad 5)$, class number 5500; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 896 & 877 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 357)\, y^3 + (38s - 114069)\, y^2 + (-635s + 4146437)\, y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 717*y^8 - 229526*y^7 + 31601333*y^6 - 1933306401*y^5
+ 77857898333*y^4 - 2073339028520*y^3 + 38405453887314*y^2 - 4438833
22121305*y + 2929811939259475
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \Big(-\frac{392}{138062368994593}\Big)\, s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12725614073817 = 3^3 \cdot 471319039771$:

$$\begin{pmatrix}
1413957119313 & 348818173482 & 1114084523112 & 1352294157834 & 342096363317 & 824465606609 & 156655225808 & 110565819356 & 1235078961937 & 496652078670 \\
0 & 3 & 0 & 0 & 2 & 0 & 1 & 2 & 1 & 1 \\
0 & 0 & 3 & 0 & 1 & 1 & 1 & 0 & 2 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 357)\, y^3 + (38s - 114069)\, y^2 + (-635s + 4146437)\, y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 717*y^8 - 229526*y^7 + 31601333*y^6 - 1933306401*y^5
+ 77857898333*y^4 - 2073339028520*y^3 + 38405453887314*y^2 - 4438833
22121305*y + 2929811939259475
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + (\frac{746}{5260658229025227})\, s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12725614073817 = 3^3 \cdot 471319039771$:

$$\begin{pmatrix} 1413957119313 & 348818173482 & 1114084523112 & 1352294157834 & 342096363317 & 824465606609 & 156655225808 & 110565819356 & 1235078961937 & 496652078670 \\ 0 & 3 & 0 & 0 & 2 & 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 3 & 0 & 1 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 357)\, y^3 + (38s - 114069)\, y^2 + (-635s + 4146437)\, y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 717*y^8 - 229526*y^7 + 31601333*y^6 - 1933306401*y^5
+ 77857898333*y^4 - 2073339028520*y^3 + 38405453887314*y^2 - 4438833
22121305*y + 2929811939259475
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right) s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9469881199371 = 3 \cdot 3156627066457$:

$$\begin{pmatrix} 9469881199371 & 7161191189476 & 7085545526483 & 8409893984811 & 9413668750897 & 6552318643806 & 1854373582889 & 4900672185672 & 7818925049375 & 2559453208712 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 801)\, y^3 + (-4s + 146024)\, y^2 + (645s - 1438566)\, y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 1617*y^8 + 285640*y^7 + 19321939*y^6 + 315860928*y^5
- 9487944165*y^4 - 541251861120*y^3 - 4379027249088*y^2 + 1050995603
49696*y + 3172815876243456
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right) s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2012190851484 = 2^2 \cdot 3^5 \cdot 11 \cdot 188195927$:

$$\begin{pmatrix} 12420931182 & 5518516710 & 1217295990 & 7244974677 & 10554051375 & 2822054093 & 10234598128 & 10778534868 & 1045305715 & 9700512824 \\ 0 & 3 & 0 & 0 & 0 & 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 3 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 150 factors with signed exponents of absolute value up to 390153445668; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 801)\, y^3 + (-4s + 146024)\, y^2 + (645s - 1438566)\, y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 1617*y^8 + 285640*y^7 + 19321939*y^6 + 315860928*y^5
- 9487944165*y^4 - 541251861120*y^3 - 4379027249088*y^2 + 1050995603
49696*y + 3172815876243456
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + (\frac{746}{5260658229025227})\, s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17320006900416 = 2^6 \cdot 3 \cdot 11^5 \cdot 560123$:

$$\begin{pmatrix} 1626597192 & 1468253204 & 150269740 & 1150889212 & 1382772184 & 713437330 & 1345522113 & 1575711433 & 1082871654 & 137899727 \\ 0 & 242 & 193 & 88 & 4 & 65 & 138 & 94 & 238 & 16 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 22 & 20 & 1 & 7 & 2 & 2 & 21 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 368821611190; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 801)\, y^3 + (-4s + 146024)\, y^2 + (645s - 1438566)\, y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 1617*y^8 + 285640*y^7 + 19321939*y^6 + 315860928*y^5
- 9487944165*y^4 - 541251861120*y^3 - 4379027249088*y^2 + 1050995603
49696*y + 3172815876243456
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + (-\frac{26359}{450359356492424224})\, s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $122784686044416 = 2^8 \cdot 3^4 \cdot 11 \cdot 17^3 \cdot 109567$:

$$\begin{pmatrix}
16719047664 & 1975027836 & 1065226170 & 2091891192 & 14106478437 & 6010757004 & 15099545454 & 2660213807 & 3069441170 & 6567327435 \\
0 & 102 & 30 & 36 & 51 & 58 & 9 & 95 & 6 & 74 \\
0 & 0 & 3 & 0 & 1 & 0 & 2 & 0 & 2 & 0 \\
0 & 0 & 0 & 24 & 13 & 17 & 7 & 14 & 8 & 13 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 12865992887; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)\, y^2 + (208s - 4320840)\, y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 686*y^8 + 62396*y^7 - 8756623*y^6 + 72623492*y^5 + 95
02752388*y^4 - 429426473408*y^3 + 9447602402368*y^2 - 99308430893568
*y + 7017616947560448
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right) s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $62418872678400 = 2^{11} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 716707$:

$$\begin{pmatrix} 19265084160 & 6029996130 & 14573151825 & 1615094022 & 15933457212 & 10530512397 & 8625296352 & 5375110570 & 12220580154 & 18233629617 \\ 0 & 30 & 15 & 24 & 0 & 14 & 17 & 28 & 13 & 10 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 6 & 0 & 3 & 2 & 4 & 5 & 2 \\ 0 & 0 & 0 & 0 & 6 & 5 & 5 & 5 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 333270424482; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)\, y^2 + (208s - 4320840)\, y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 686*y^8 + 62396*y^7 - 8756623*y^6 + 72623492*y^5 + 95
02752388*y^4 - 429426473408*y^3 + 9447602402368*y^2 - 99308430893568
*y + 7017616947560448
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227}\right) s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13461303389184 = 2^{10} \cdot 3^3 \cdot 7 \cdot 69554519$:

$$\begin{pmatrix} 17527738788 & 16275929212 & 537995854 & 14471558808 & 11874756348 & 1241162132 & 6962415374 & 533604480 & 12029995317 & 10675394590 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 6 & 2 & 2 & 4 & 4 & 1 & 5 \\ 0 & 0 & 0 & 0 & 4 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 221 factors with signed exponents of absolute value up to 495178176932; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 3 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)\, y^2 + (208s - 4320840)\, y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 686*y^8 + 62396*y^7 - 8756623*y^6 + 72623492*y^5 + 95
02752388*y^4 - 429426473408*y^3 + 9447602402368*y^2 - 99308430893568
*y + 7017616947560448
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right) s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $622375914674304 = 2^7 \cdot 3^4 \cdot 7 \cdot 8575505879$:

$$\begin{pmatrix} 1440684987672 & 34252957320 & 344180541252 & 1430544082658 & 665874173682 & 803579986447 & 728210031805 & 506178240890 & 168047728331 & 354151960872 \\ 0 & 12 & 0 & 3 & 0 & 5 & 8 & 10 & 7 & 7 \\ 0 & 0 & 12 & 4 & 3 & 1 & 7 & 10 & 4 & 7 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 6694573388851; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 3401y^3 + (-45s - 61044)\,y^2 + (465s - 4980174)\,y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 6802*y^8 - 122133*y^7 + 1606918*y^6 - 320021187*y^5 + 1587530
3883*y^4 - 18563558472*y^3 - 3500133396633*y^2 - 191813227477479*y +
6348829617810441
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right) s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $324410514353559 = 3^8 \cdot 41 \cdot 24097 \cdot 50047$:

$$\begin{pmatrix}
445007564271 & 308680969104 & 249262503192 & 432378669465 & 8813331765 & 259193366509 & 383759122093 & 116393320134 & 267547266960 & 43868885602 \\
0 & 9 & 0 & 3 & 0 & 1 & 6 & 8 & 2 & 0 \\
0 & 0 & 9 & 0 & 0 & 1 & 0 & 2 & 3 & 0 \\
0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 1 & 1 \\
0 & 0 & 0 & 0 & 3 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 566 factors with signed exponents of absolute value up to 7882758077971264855514; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 3401y^3 + (-45s - 61044)\,y^2 + (465s - 4980174)\,y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 6802*y^8 - 122133*y^7 + 1606918*y^6 - 320021187*y^5 + 1587530
3883*y^4 - 18563558472*y^3 - 3500133396633*y^2 - 191813227477479*y +
6348829617810441
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \Big(\frac{746}{5260658229025227}\Big)\,s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $35966359904463 = 3^5 \cdot 461 \cdot 1097 \cdot 292673$:

$$\begin{pmatrix}
444029134623 & 168034694727 & 410738647212 & 828149637 & 81325220199 & 303226736764 & 7097437813 & 229662447121 & 367474967190 & 162334466191 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 3 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 577 factors with signed exponents of absolute value up to 920874362386795255491; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 3401y^3 + (-45s - 61044)\,y^2 + (465s - 4980174)\,y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 6802*y^8 - 122133*y^7 + 1606918*y^6 - 320021187*y^5 + 1587530
3883*y^4 - 18563558472*y^3 - 3500133396633*y^2 - 191813227477479*y +
6348829617810441
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + (-\frac{26359}{450359356492424224})\,s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25350282685575 = 3^3 \cdot 5^2 \cdot 39293 \cdot 955793$:

$$\begin{pmatrix}
563339615235 & 31909046205 & 428426494329 & 255401352753 & 375637722716 & 494008992530 & 136544952316 & 270432280855 & 105047096078 & 198896620463 \\
0 & 5 & 3 & 2 & 4 & 2 & 3 & 0 & 0 & 1 \\
0 & 0 & 3 & 0 & 2 & 2 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 3 & 2 & 1 & 0 & 0 & 2 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 262)\, y^3 + (24s + 129615)\, y^2 + (75s + 1637544)\, y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 529*y^8 + 258204*y^7 + 25551363*y^6 + 1166990841*y^5
+ 34132524471*y^4 + 602656235370*y^3 + 6927113426958*y^2 + 309217678
17327*y + 147664883721339
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + (-\frac{392}{138062368994593})\, s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $901177749216 = 2^5 \cdot 3^2 \cdot 2081 \cdot 1503647$:

$$\begin{pmatrix}
18774536442 & 18059618250 & 17063296464 & 12299518994 & 12619287786 & 8081837136 & 7294084140 & 747208492 & 3874020284 & 669837582 \\
0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\
0 & 0 & 6 & 2 & 0 & 3 & 0 & 1 & 5 & 4 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 262)\, y^3 + (24s + 129615)\, y^2 + (75s + 1637544)\, y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 529*y^8 + 258204*y^7 + 25551363*y^6 + 1166990841*y^5
+ 34132524471*y^4 + 602656235370*y^3 + 6927113426958*y^2 + 309217678
17327*y + 147664883721339
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \Big(\frac{746}{5260658229025227}\Big)\, s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $901177749216 = 2^5 \cdot 3^2 \cdot 2081 \cdot 1503647$:

$$\begin{pmatrix}
18774536442 & 18059618250 & 17063296464 & 12299518994 & 12619287786 & 8081837136 & 7294084140 & 747208492 & 3874020284 & 669837582 \\
0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\
0 & 0 & 6 & 2 & 0 & 3 & 0 & 1 & 5 & 4 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 262)\, y^3 + (24s + 129615)\, y^2 + (75s + 1637544)\, y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 529*y^8 + 258204*y^7 + 25551363*y^6 + 1166990841*y^5
+ 34132524471*y^4 + 602656235370*y^3 + 6927113426958*y^2 + 309217678
17327*y + 147664883721339
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right) s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21406235190729 = 3 \cdot 17^2 \cdot 24690005987$:

$$\begin{pmatrix}
1259190305337 & 771631666323 & 817612348020 & 1004432335548 & 730042912528 & 874207267467 & 1070663323203 & 894740804307 & 53338463781 & 548546532485 \\
0 & 17 & 1 & 13 & 4 & 13 & 4 & 2 & 6 & 3 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 602)\, y^3 + (5s - 115482)\, y^2 + (220s + 636744)\, y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1204*y^8 - 232162*y^7 + 24592271*y^6 - 348318940*y^5
+ 4651307932*y^4 - 110306595104*y^3 - 580516628672*y^2 + 17625609739
776*y + 98258841772032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + (-\frac{392}{138062368994593})\, s, \qquad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57982493392896 = 2^{18} \cdot 3 \cdot 73728553$:

$$\begin{pmatrix}
7077941088 & 7031309552 & 5057279344 & 4006051400 & 2736104252 & 6719377740 & 6521232414 & 3572812626 & 5409477834 & 3140639326 \\
0 & 16 & 0 & 10 & 14 & 2 & 14 & 15 & 12 & 5 \\
0 & 0 & 16 & 14 & 0 & 2 & 2 & 10 & 10 & 9 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 346 factors with signed exponents of absolute value up to 44819004878; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 602)\, y^3 + (5s - 115482)\, y^2 + (220s + 636744)\, y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1204*y^8 - 232162*y^7 + 24592271*y^6 - 348318940*y^5
+ 4651307932*y^4 - 110306595104*y^3 - 580516628672*y^2 + 17625609739
776*y + 98258841772032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + (\frac{746}{5260658229025227})\, s, \qquad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65102805882112 = 2^8 \cdot 35753 \cdot 7112909$:

$$\begin{pmatrix}
1017231341908 & 803902129056 & 496726163974 & 736394910314 & 94518748456 & 916198534922 & 286853387180 & 813221468902 & 133898644251 & 95777554658 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 379 factors with signed exponents of absolute value up to 148486256644; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 602)\, y^3 + (5s - 115482)\, y^2 + (220s + 636744)\, y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1204*y^8 - 232162*y^7 + 24592271*y^6 - 348318940*y^5
+ 4651307932*y^4 - 110306595104*y^3 - 580516628672*y^2 + 17625609739
776*y + 98258841772032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + (-\frac{26359}{450359356492424224})\, s, \qquad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65102805882112 = 2^8 \cdot 35753 \cdot 7112909$:

$$\begin{pmatrix} 1017231341908 & 803902129056 & 496726163974 & 736394910314 & 94518748456 & 916198534922 & 286853387180 & 813221468902 & 133898644251 & 95777554658 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 351 factors with signed exponents of absolute value up to 60778112507; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.